

OPEN-PIT MINES

Some of the largest mines are open holes in the ground, dug by colossal mechanical excavators. Huge trucks drive down access roads to the bottom of the mine to be loaded up. Open-pit mines are used for coal close to the surface, but the deepest pits are mines for metals such as copper and the radioactive uranium used as fuel in nuclear power plants.

OPEN PIT ►

A truck rolls up the terraces of an open-pit mine, carrying 250 tons of rock. When the mine is exhausted, it will become a waste landfill.



DEEP SHAFTS

The most expensive and dangerous form of mining involves tunneling deep below ground. Horizontal galleries are linked to vertical shafts, which have lifts that carry miners down to seams. Deep-shaft mines have to be constantly drained of water and cooled, and miners such as these men in South Africa risk rockfalls and gas explosions.



FAST FACTS

- The deepest mine is the TauTona gold mine in South Africa, which extends 2.4 miles (3.9 km) below the surface.
- The deepest open-pit mine is the Bingham Canyon copper mine in Utah—0.75 miles (1.2 km) deep, it is the biggest artificial hole on Earth.
- In 2009, an offshore oil well in the Gulf of Mexico was drilled to a depth of 6.6 miles (10.7 km).



DRILLING FOR OIL

By drilling boreholes down through rock, miners can reach reserves of liquid oil and natural gas. Deep beneath the ground, these fluids are often under pressure that squeezes them up the borehole to the surface. Offshore drilling platforms tap reserves beneath the seabed. This rig is drilling off the coast of California but some are towed out to far deeper waters.